FIFA World Cup 2026 Predictor

Academic methodology brief — training data & methods (for methodological review)
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> **Summary.** We train on *international A-team matches only* (country vs country), using a **5-year rolling window** before each World Cup cutoff. Recent matches are **upweighted** (2-year half-life decay), not excluded. The model does **not** use club football (Premier League, Champions League, etc.).

DATA SOURCES

- **Historical training:** Kaggle martj42 international results (`results.csv`) — A-team match outcomes from 1872 onward.
- **2026 fixtures:** FIFA.com official schedule/PDF (`2026_FIFA_World_Cup_Schedule.csv.csv` locally), with playoff slots resolved from UEFA/FIFA playoff results.
- **DataCamp files:** used only for competition *submission format* (match_id alignment, output schema, placeholder team labels for grading) — not authoritative for kickoff times, qualification, or real-world truth.
- **Secondary checks (optional):** OpenFootball or TheStatsAPI as machine-readable cross-checks against the FIFA schedule.

WHAT MATCHES ARE INCLUDED?

Training rows come from the Kaggle martj42 dataset above, extended through 2026. Teams are **nations** (France, England, Brazil) — never clubs.

- **Included:** World Cup & continental qualifiers, continental finals (Euro, Copa, AFCON, Asian Cup), Nations League, Gold Cup, friendlies, and regional A-team cups — all confederations (UEFA, CONMEBOL, CAF, AFC, CONCACAF, OFC).
- **Excluded:** Domestic leagues, UEFA Champions League/Europa League, any club match; matches on/after the forecast cutoff; matches older than 5 years.
- **Validation:** Walk-forward — tune on 2018 World Cup (cutoff opening day), confirm on 2022; 64 finals matches held out per year for Brier scoring.

TRAINING WINDOWS (VERIFIED COUNTS)

Run	Cutoff	Matches	From	To
2018 development	2018-06-14	4,617	2013-06-15	2018-06-12
2022 confirmation	2022-11-20	4,473	2017-11-22	2022-11-19

2026 production	2026-06-11	5,239	2021-06-12	2026-06-05

Each run uses only the last 1825 days (~5 years) with `date < cutoff`. The full archive spans 1872–2026 (~49k matches); training never uses the entire history.

CLUB VS NATIONAL TEAM — COMMON CLARIFICATION

NOT IN OUR DATA

PSG → France, Arsenal → England (club → nation mapping)
Premier League, Ligue 1, Champions League
Club form / domestic league progression

WHAT WE ACTUALLY USE

France vs Brazil, England vs Germany (A-team fixtures)
UEFA = *national-team* Euro, qualifiers, Nations League
Strength learned from weighted international results + Elo

HOW MATCHES ARE WEIGHTED (ITERATION 7)

Component	Setting	Meaning
Time decay	2-year half-life	Matches in the last 2 years get the highest weight; older matches within the 5-year window count less.
Tournament tier	A=1.00, B=0.85, C=0.50	WC/qualifiers/finals (A); Nations League, Gold Cup (B); friendlies (C).
Opponent difficulty	D_{opp}	Discounts goals vs weak opponents (walk-forward Elo).
Competition context	C_{conf} (Opta-inspired)	FIFA 1.05, UEFA 1.00, CONMEBOL 0.95, CAF 0.88, AFC 0.80, CONCACAF 0.78 — discounts weaker competition *contexts*, not teams.

Final sample weight = time × tournament tier × D_{opp} × C_{conf}. Model: Dixon-Coles Poisson with Elo anchor; neutral venues at World Cup inference.

WHY MONTE CARLO SIMULATION?

Dixon-Coles Poisson yields **match-level** score probabilities — not tournament-stage outcomes directly. The 2026 format adds structural complexity: 12 groups of four, eight best-third-place qualifiers assigned to predetermined knockout slots, and a 32-team bracket where later opponents depend on earlier group results.

- **Bracket dependencies:** Knockout pairings follow group standings and best-third tiebreak rules; a closed-form tournament-winner probability is not available.
- **Quantities estimated:** P(win), P(reach semifinals), group advancement rates, and uncertainty in best-third qualification.

- **Method:** Many independent simulated tournaments (10,000 in production) sample match scores from the Poisson model, apply FIFA tiebreak rules, and aggregate stage outcomes.

OPTA-INSPIRED WEIGHTS VS OPTA BENCHMARK

OPTA-INSPIRED (IN TRAINING)

C_{conf} multipliers on each *match* by tournament confederation context.
Tuned on 2018 Brier. Does **not** import OPTA team ratings.

OPTA CSV (BENCHMARK ONLY)

Pre-tournament win % from OPTA used only for post-hoc comparison tables — never a training feature.

BACKTEST PERFORMANCE (BRIER — LOWER IS BETTER)

Year	Iteration 7 Brier	Role
2018	0.739	Development / hyperparameter tuning
2022	0.726	Holdout confirmation

ITERATION 8 (PROTOTYPE)

An exploratory branch blends Transfermarkt squad-value priors with walk-forward Elo before Poisson training. Production forecasts remain on iteration 7; iteration 8 is retained for sensitivity analysis only.